



Safer Running of Processes and Risk Minimisation in the Operation of Plants and Machines

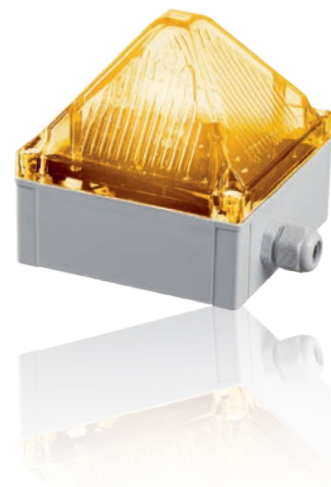
Safety-related Sounders and Flashing Lights

Edition 15

Functional Safety IEC 61508 | IEC 61511 (SIL)



SOUNDER
DS SERIES



FLASHING LIGHT
Quadro SERIES



ALL-ROUND LIGHT
PMF SERIES



Functional Safety EN ISO 13849-1 (PL)

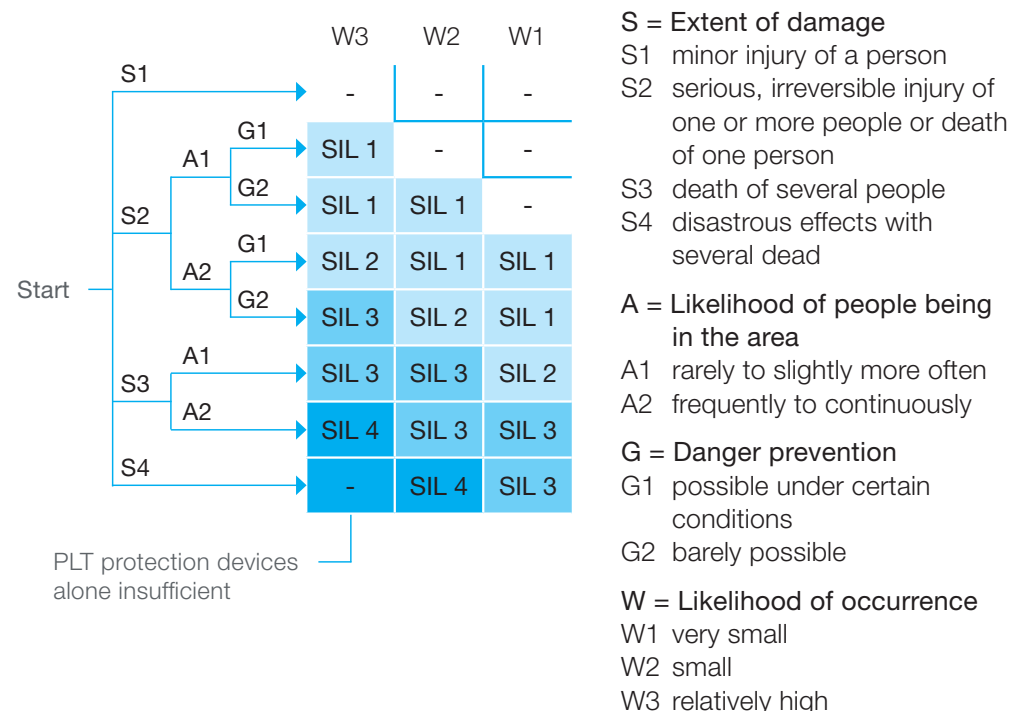
Since 1st June 2015, the European Directive, Seveso III, has been implemented as national law. In Germany, this occurs with the Amendment to the Hazardous Incident Ordinance (12 BImSchV). Official, government inspections and monitoring will have even more influence in the new Hazardous Incident Ordinance.

Basic obligations are to take precautions to prevent hazardous incidents and the possible consequences thereof. This includes equipping the operating area with sufficient warning, alerting and safety facilities. For many years Pfannenberg has provided SIL/PL compliant signalling technology for harsh industrial environments. Equipped with the key safety data, the sounders and flashing lights can be integrated in the safety concept of machines and plants without any problems.

Machinery directive 2006/42/EG has been valid since the 1st January 2010. It was signed on 17 May 2006 and published in the Official Journal of the European Union (OJ L 157) on 9 June 2006. Two safety standards became effective with the machinery directive. On the one hand, DIN EN ISO 13849-1, which replaces Norm DIN EN 954-1 of the old machinery directive 98/37/EG. On the other hand DIN EN 62061.

Objective of these safety norms is to minimize risks in the operation of machines. Therefore, the requirements for the certification of products for machine and plant manufacturers were increased. Probability considerations, amongst other things, have an impact on the regulation of the safety of components. In order to keep the existing residual risk of a machine or plant as low as possible, alarm devices which have a high functional safety are required. They should also alert about danger via visual or audible warning signals.

HAZARD GRAPH ACCORDING TO IEC 61508



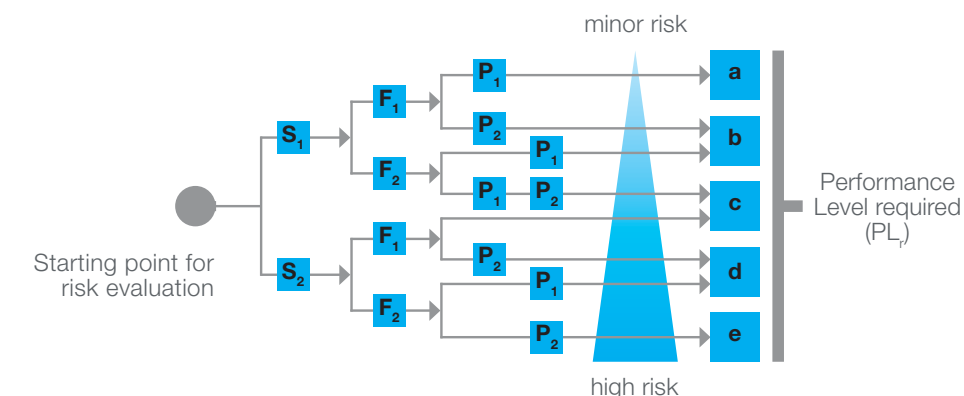
SAFETY INSTRUMENTED SYSTEM SIS (SAFETY LOOP)



As a rule, alarm equipment performs a safety protection function on machines and systems. Therefore, the consequences of an error in the signaling devices always present a potential risk if not seen.

A hazard graph is not only an important element for the classification of the system or plant, it also clearly illustrates how complex the preliminary work is for the realisation and implementation of projects.

RISK ASSESSMENT



Determination of the required Performance Levels (PL_r)

RISK PARAMETERS

- **S – Severity of the injury**
S₁ = minor injuries (normally reversible)
S₂ = serious injuries, including death (normally irreversible)
- **F – Frequency and/or length of exposure to the hazard**
F₁ = rarely to more often and/or short periods
F₂ = frequently to continuously and/or long duration
- **P – Possibilities to avoid danger**
P₁ = possible under certain conditions
P₂ = barely possible

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